UNIVERSITY OF OTAGO EXAMINATIONS 2017

COMPUTER SCIENCE

COSC343

ARTIFICIAL INTELLIGENCE

Semester 1

(TIME ALLOWED: THREE HOURS)

This examination comprises 7 pages.

Candidates should answer questions as follows:

Candidates should answer **5** questions.

Questions are worth 12 marks and sub marks are shown thus: (2)

The total number of marks for this exam is 60.

The following material is provided:

Nil.

Use of calculators

No restriction on the model of calculator that may be used, but no device with communication capability shall be accepted as a calculator. Calculators are subject to inspection by the examiners.

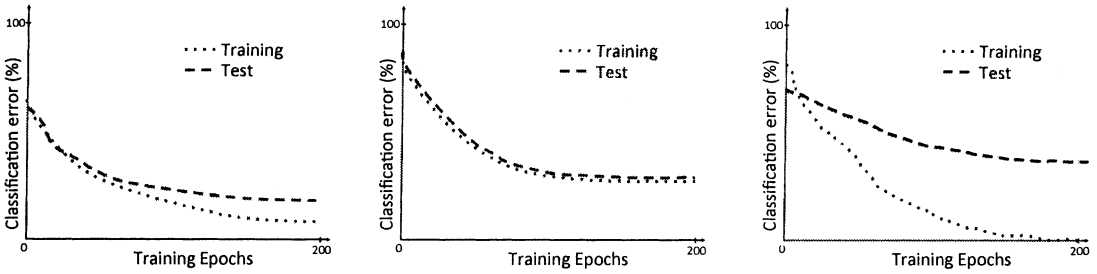
Candidates are permitted copies of:

Nil.

TURN OVER

1. Learning

The figure below shows the classification error as a function of training epochs on the same set of training and test data using three different hypotheses (which you can refer to as hypothesis $H1$, $H2$ and $H3$).



$H1$ $H2$ $H3$

(a) Explain what it means for a hypothesis to be *consistent*. Based on the training and the test error shown above, which hypothesis is the most and which one is the least *consistent* on the last epoch shown? (3)

(b) Explain what it means for a hypothesis to *overfit* the data? Which one of the hypotheses above *overfits* the data the most and which one the least? (3)

(c) Explain what it means for a hypothesis to *generalise*? Which one of the hypothesis seems to generalise best, and which one appears to give the worst generalisation? (3)

(d) If you were to pick one hypothesis from the above, which one would you choose? Why? (3)

2. **Uncertainty**

Consider the following probability distribution for three boolean random variables: $Toothache$, $Catch$ and $Cavity$.

	$toothache$		$\neg toothache$	
	$catch$	$\neg catch$	$catch$	$\neg catch$
$cavity$	0.1	0.01	0.04	0.01
$\neg cavity$	0.01		0.23	0.5

Recall that:

$$P(a, b) = P(a|b)P(b) = P(b|a)P(a)$$

(a) What is the missing value for $P(toothache, \neg cavity, \neg catch)$? Show how you computed it. (1)

(b) Compute $P(toothache|cavity)$ and $P(\neg catch|cavity)$. (4)

(c) Let's say there is a patient who reports toothache but the probe doesn't catch. What diagnosis with respect to presence of a cavity would a Naive Bayes Classifier give? Show your calculations. (5)

(d) If you were to construct a simple decision tree to perform a diagnosis, how would you determine which symptom should the tree examine first: $Toothache$ or $Catch$? Describe how this can be done and the probabilities involved. There is no need to compute the answer, just explain the process that determines the answer. (2)

3. Neural networks and deep learning

Tables $T1$ and $T2$ below define desired output y for combinations of two-attribute inputs x_1 and x_2.

x_1	x_2	y
0	0	1
0	1	1
1	0	1
1	1	0

$T1$

x_1	x_2	y
0	0	0
0	1	1
1	0	1
1	1	0

$T2$

(a) Consider tables $T1$ and $T2$ above, which define the desired output y for combinations of two-attribute inputs x_1 and x_2. In each case, is a *perceptron* capable of consistent classification of the data? Explain why or why not. (2)

(b) Explain the behaviour of the learning rule over an arbitrary number of epochs for $T1$ as opposed to $T2$. Think in terms of convergence. (2)

(c) What is a *multilayer feed-forward neural network*? Describe all the elements that the model is composed of and how it can be trained in a *supervised* learning scenario. (3)

(d) What would be different in a neural network architecture and training if you were to set it up for a *classification* as opposed to a *regression* problem? (2)

(e) How does a *convolutional neural network* relate to a multilayer feed forward neural network, and why is it better suited for finding patterns in images? (3)

4. **Unsupervised learning**

Let's say there is a space-probe that sends over PCA compressed data from its array of sensors. One day the probe sends over a 32×548 matrix (32 rows and 548 columns) of data which corresponds to several measurements over the sensor array stored over a number of days. You have a copy of the 32×660 PCA encoding matrix.

(a) How do you decode the data? Explain how the decoding works. (3)

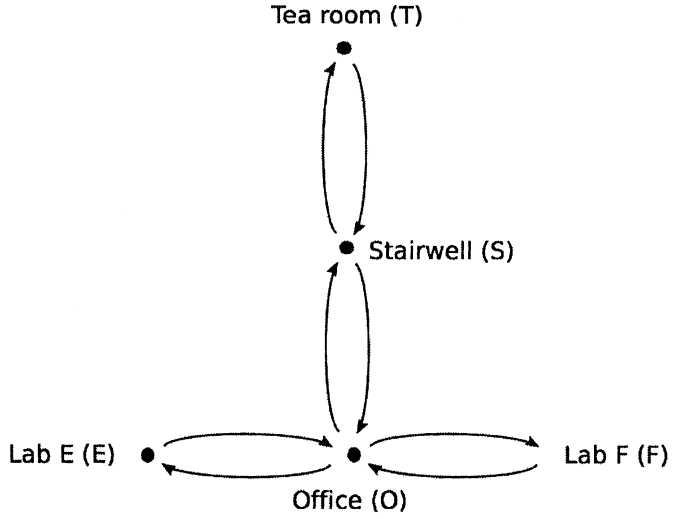
(b) How many measurements are there in the dataset, and how many sensors are there on the space-probe? Justify your answer. (3)

(c) Is the compression lossless (restored data is exactly same as the original) or lossy (restored data is an approximation of the original)? Explain. (2)

(d) Scientists are interested in finding some patterns in the received data, but they don't know what possible patterns there could be. They would like to start by finding out whether the data inherently lends itself to classification by unknown categories. What unsupervised learning technique could you use? If you had to explain to the scientists how it works and what the limits of the technique are, what would you say? (4)

5. Search

An automaton is disguised as a Computer Science student, and has infiltrated the Owheo building. The automaton has a state space graph representing the important places in the building, as shown below.



The nodes in the graph represent states the automaton can be in. (Abbreviations are shown in brackets.) The directed arcs represent the actions it can take.

(a) Assume the automaton is at Lab E (E), and wants to go to the tea room (T). Write down the **search tree** the automoton would generate if it was searching the state space using **breadth-first search**. Assume that the local ordering of a parent node's children is determined by alphabetical order of abbreviations of the associated states: thus if a node's children denote the tea room (T) and the office (O), they will be added to the fringe in the order (O, T). (3)

(b) Explain why a simple **depth-first search** would be problematic in this state space. What extension to depth-first search would address this problem? (2)

(c) What is the **iterative deepening search** algorithm? Explain how this algorithm combines the advantages of depth-first and breadth-first search, referring to **optimality**, **space complexity** and **time complexity**. Explain in particular why iterative deepening has better time complexity than breadth-first search. (4)

(d) Assume that the above state space graph is just a fragment of the full graph used by the automaton. Suggest a heuristic search algorithm, and a heuristic, that will make path finding tractable in this larger domain. Illustrate your answer in relation to the state space fragment you were given. (3)

6. Natural Language

Consider the following sentence:

Alice hit the teacher on the nose.

And the following context-free grammar:

S	→	NP, VP	PN	→	Alice
NP	→	PN	V1	→	hit
NP	→	DET, N	DET	→	the
VP	→	V1, NP	N	→	teacher
PP	→	???	N	→	nose
VP	→	???	P	→	on
NP	→	???			

(a) There is a **syntactic ambiguity** in the sentence, relating to the phrase 'on the nose', which can be interpreted in two different ways. Write down a sentence that clearly expresses each possible interpretation of the sentence. (2)

(b) Three of the rules in the grammar are incomplete, as shown by the '???' expressions.

(i) Write down completions of these three rules, so that the complete rules allow two analyses of the ambiguous sentence. (3)

(ii) Write down the two alternative syntactic analyses of the ambiguous sentence, indicating which analysis corresponds to which interpretation. (2)

(c) Describe how a **statistical grammar** could be developed, that would allow resolution of this ambiguity. Your answer should explain what a statistical grammar is, how it is trained, and how it can be used to resolve syntactic ambiguities. (5)

END